

Book Content & Beliefs Survey

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Author Note

The following is from an RIT statistics club Hackathon using data from a Columbia study titled “Quantifying Emotion”. Participants worked alone during a consecutive 4 hour timeframe to find any interesting insights in this dataset. Pages 2-5 (4 pages) encompass the report, while pages 7-14 (appendix) contain supporting visuals.

Book Content & Beliefs Survey

The data is from a study titled “Quantifying Emotion” run by Professor Hortense Fong at Columbia University. The goal of the study was to understand reader story expectations. Participants read two chapters from each of two fictional stories, then rated their emotional reactions—measured using valence and arousal—for each chapter. Valence measures how positive or negative the participant feels and arousal measures how high energy or low energy the participant feels. The data includes additional behavioral data, attention checks, genre preference questions, and free-text responses. A summary of columns is found in Appendix A1.

A Python notebook and JMP workbook (analogous to a STATA workbook) were used for this assignment and are available for download¹ and LaTeX was used to formalize the report. Throughout the analysis, I consulted online forums like StackExchange and chatbots like ChapGPT to help brainstorm and debug code. This assignment was completed over several days in approximately 8 hours in total, broken down in Appendix A1.

Data Preprocessing

The data was relatively complete and needed minimal cleaning. Rows 6 and 200 were combined into a single response because they contain each other’s complementary information. This was most likely a single response that was broken up into two responses by error. Row 7 contained an “I” in a numeric column for which I inferred it to be a 1. Row 1 contained all survey prompts, which I transferred to a different file to keep as a reference, creating keys to match the now separate datasets. A handful of computed columns were created to support analysis and described in Appendix A1.

Analysis

To start, I generated summary statistics for the primary observations valence and arousal, and looked at a correlation map to see if anything stuck out. At the same time I uploaded assignment in ChatGPT to get recommendations on where to begin (A2). Considering the feedback, limited time, and summary output, I decided to focus on certainty ratings. By focusing on this variable we can begin to understand how users determine confidence in story expectations.

¹ https://github.com/erf-git/EDA_Book_Beliefs

Figure 1 portrays participant certainty scores declining as the survey progresses in both valence and arousal hinting at a possible decline in user attention. My initial thought was that the chapters were the culprit of the decline. Potentially certain texts or genre preferences were less engaging for the participants. To investigate this, I recorded the correlations of certainty between genre, chapters (1-hot encoded), and other user metrics as seen in Figure 2.

Figure 1

Plots of diminishing valence and arousal certainty over chapters. Certainty rating decreases from chapter 1 to chapter 2 and overall decreases throughout the survey.

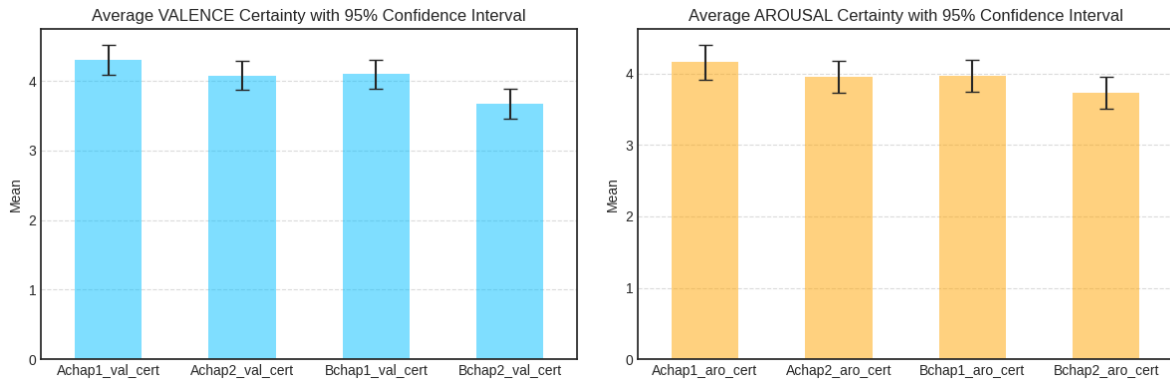
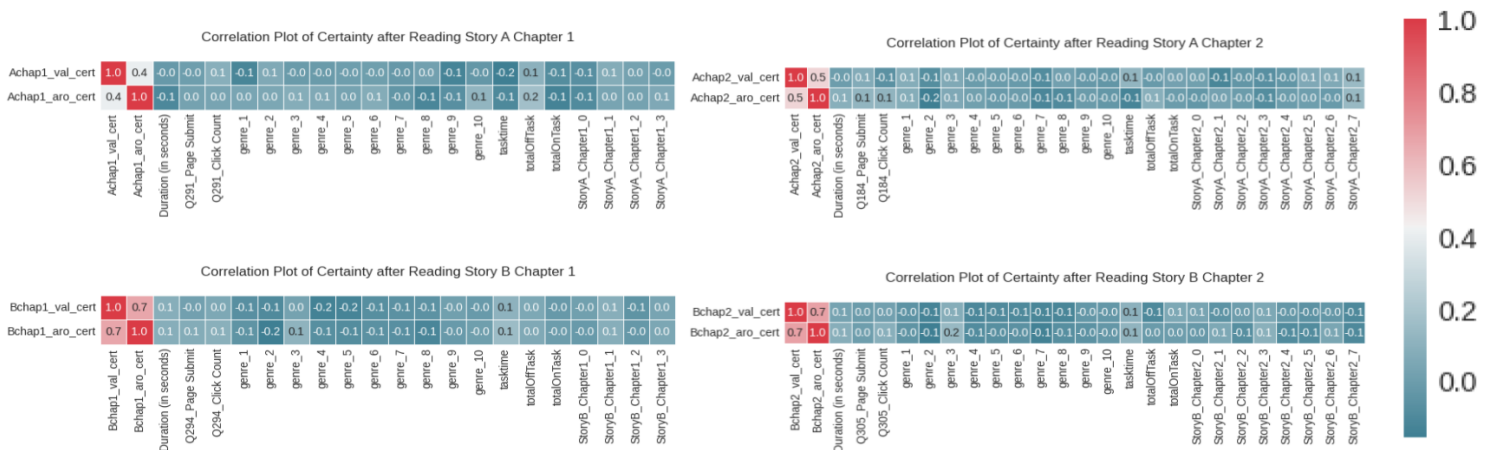


Figure 2

Correlation heatmaps for valence and arousal certainty for each chapter.



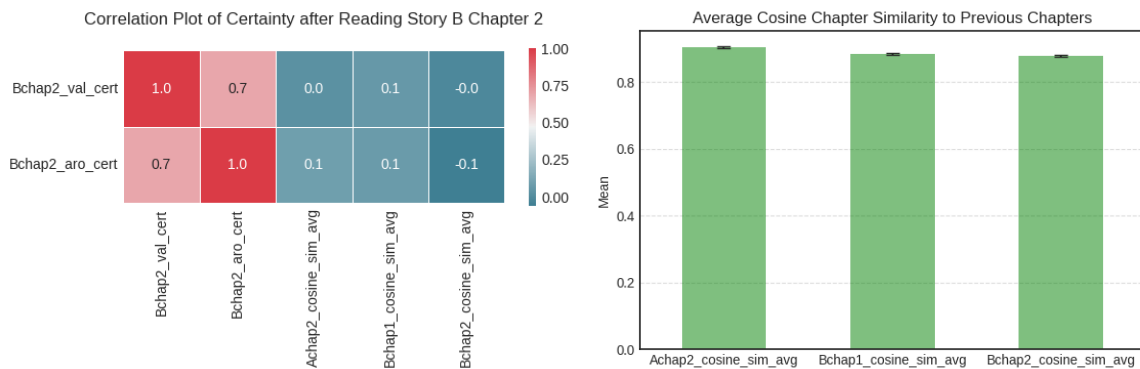
Initially, we see valence and arousal certainty ratings are highly correlated with each other. This is intuitive, but not an interesting finding. There is slight evidence for a couple of correlations: A_Chapter1 arousal and total time distracted (offTaskTime), A_Chapter1 valence

and total time spent on the key task (taskTime), A_Chapter2 or B_Chapter1 arousal and number of fantasy books read (genre_2), B_Chapter1 valence and number of mystery or thriller books read (genre_4 and genre_5), B_Chapter2 arousal and number of biographies read (genre_3). However, nothing is consistent or strongly correlated. If given more time, exploration of how genre influences audiences might be a valuable point of inquiry.

To potentially rule out the effect of the text, I wanted to see if the chapters were similar to previous ones. All participants read the same chapters and stories in the same order (Chapter 1 then 2 of stories A then B). The hypothesis is that if chapters become progressively dissimilar to each other, then this could cause confusion leading to drop in certainty. Chapters were transformed into embeddings using a BERT tokenizer, and cosine similarity scores between current and previous chapter were calculated. Figure 3 shows that progressively throughout the survey the average similarity between the chapters decreases, but unfortunately, there is not enough evidence to support a link between the cosine similarities and certainty in valence or arousal. A more sophisticated LLM model might be better suited to understand this phenomenon if allotted.

Figure 3

Diminishing cosine similarity scores and correlation heatmap of similarity scores by story B chapter 2.



I liked the idea that previous information could influence user certainty. We have current and previous scores for self-ratings and surprise ratings. For each chapter, I plotted the correlations between the valence and arousal certain to the current and previous valence/arousal ratings, all of which are found in Appendix A5. Looking at the correlation plot for our last chapter (Figure 4) I finally see evidence for an interesting result. The plot suggest that participants are

Figure 4

Correlation Plot of Certainty after Reading Story B Chapter 2



Given 4 more hours, I'd explore: (1) applying a machine learning model with explainability, (2) studying the influence of user attention, (3) the effect of genre on surprise. Additionally, I'd design a follow-up survey to test whether certainty diminishes over time during a longer story. Possibly I could explore disengagement/fatigue which is common in longer studies, or find an opposite effect where users certainty might increase after a percentage threshold of text is read.

References

- Folk, C., & Gibson, B. (2001). *Attraction, distraction and action: Multiple perspectives on attentional capture* (Vol. 133). Elsevier.
- Knight, S., Rocklage, M. D., & Bart, Y. (2024). Narrative reversals and story success. *Science Advances*, 10(34), eadl2013.
- Pieters, R., & Wedel, M. (2004). Attention capture and transfer in advertising: Brand, pictorial, and text-size effects. *Journal of marketing*, 68(2), 36–50.

Appendix

Supporting Figures and Tables

Figure A1
Breakdown for 4 hours spent on this report. Allowances are not exact and frequently overlap.

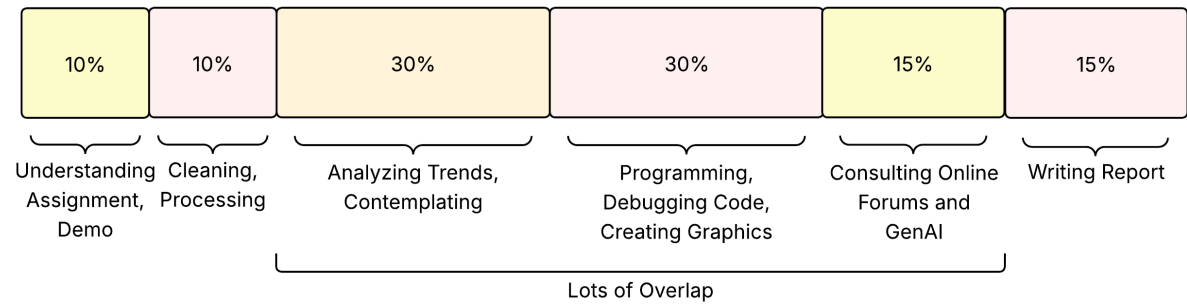


Figure A2

ChatGPT recommendations for prominent variables.

Out of these variables, which might be insightful for understanding user story expectations?

Happy to help! Could you share the list of variables you're referring to? Once I see them, I can suggest which ones are most insightful for understanding user story expectations.

Survey Variable Documentation

Metadata and Timing

StartDate: Timestamp when the respondent started the survey.

EndDate: Timestamp when the respondent ended the survey.

Duration (in seconds): Total duration of the survey in seconds.

RecordedDate: Timestamp when the response was recorded.

RespondId: Unique respondent ID.

Page Interaction Timing

Q291_First Click: ↓ click time for the initial page.

Q291_Last Click: Last click time for the initial page.

 **Belief-Based Variables (Direct Expectations)**

These are *explicit indicators* of user expectations:

- `Achap1_belief`, `Achap2_belief`, `Bchap1_belief`, `Bchap2_belief`:
Open-ended beliefs about what will happen next—**direct reflections of narrative expectations**.

 **Affective Appraisal Variables (Emotional Framing of Expectations)**

These reveal **emotional tone** of expectations and confidence:

- `*_val_scale`, `*_aro_scale`:
Valence and arousal ratings—helpful to know how users feel the story is progressing.
- `*_val_cert`, `*_aro_cert`:
Confidence in their ratings shows **how certain** they are in interpreting the story tone, which ties to expectations.
- `*_val_cert_exp1`, `*_aro_cert_exp1`:
Their explanations give deeper **rationale** for expectations.

 **Surprise Variables (Expectation Violation)**

Useful to assess **how expectations were broken**:

- `Achap2_val_surprise`, `Achap2_aro_surprise`, `Bchap2_val_surprise`, `Bchap2_aro_surprise`
— and their corresponding `*_exp1` variables:
These gauge the **gap between expectation and outcome**.

↓

Table A1

Column summary of data collected in the Quantifying Emotion study. Calculated columns are shown at the bottom with accompanying description.

Data Category	Description
Metadata and Timing	Timestamps for when the user started and ended the survey, total duration, and response ID.
Page Interaction Timing	Timestamps for when the user clicked through the pages of the survey.
Story A - Chapter 1 Responses	Valence and arousal ratings after reading chapter 1, predictions for next chapter, and rating reasoning.
Story A - Chapter 2 Responses	Valence and arousal ratings after reading chapter 2, surprise rating, predictions for next chapter, and rating reasoning.
Story B - Chapter 1 Responses	Valence and arousal ratings after reading chapter 1, predictions for next chapter, and rating reasoning.
Story B - Chapter 2 Responses	Valence and arousal ratings after reading chapter 2, surprise rating, predictions for next chapter, and rating reasoning.
Genre Preferences	Number of books the participant has read in various genres in past 10 years.
Attention and Comments	Typed responses from participants summarizing the stories read and opportunity to provide feedback.
Stimuli Text	Text from both stories.
Engagement Metrics	Time spent on and off the survey, total time spent on key tasks, and timestamps when the user switched off task.
Calculated Variable	Description
Averages	Mean rating, certainty, and surprise for valence and arousal.
Prediction Difference	Valence and arousal differences between prediction and actual, average differences.
times_distracted	Count of times distracted (off task).
Chapter Encoding	1-Hot encoding and BERT vector embedding representations of chapters.

Figure A3

ANOVA reports for valence certainty over chapter. The left report contains a means comparison over all chapters. The right report contains the mean comparison blocking for story. Both highlight a decline in user valence certainty.

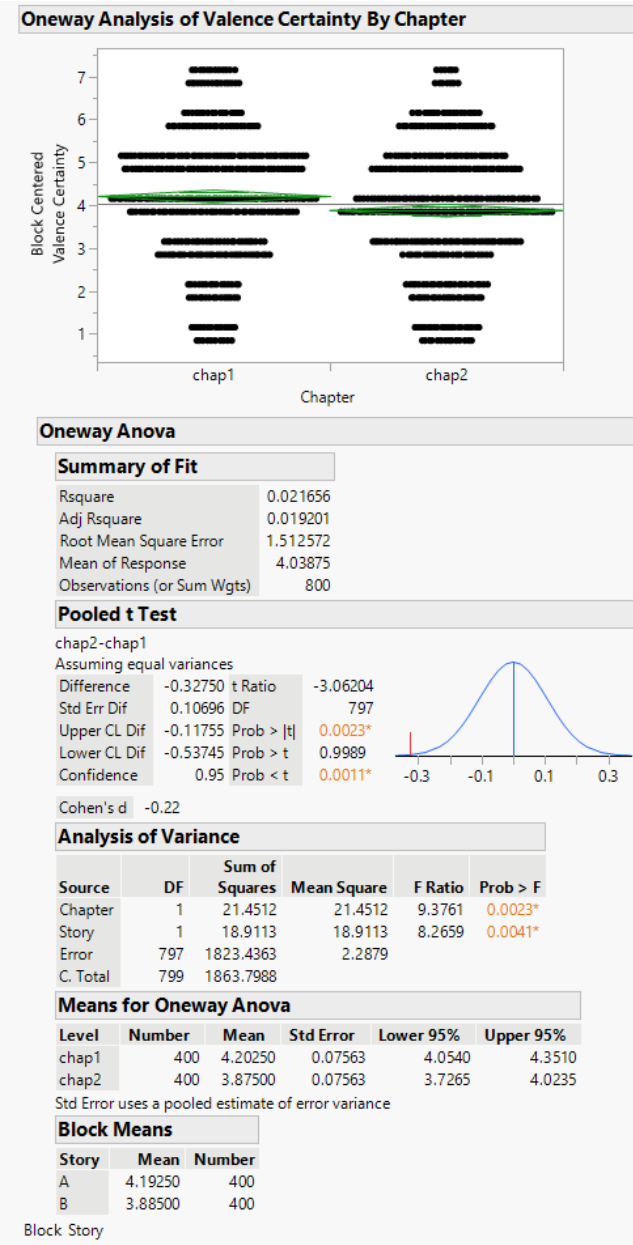
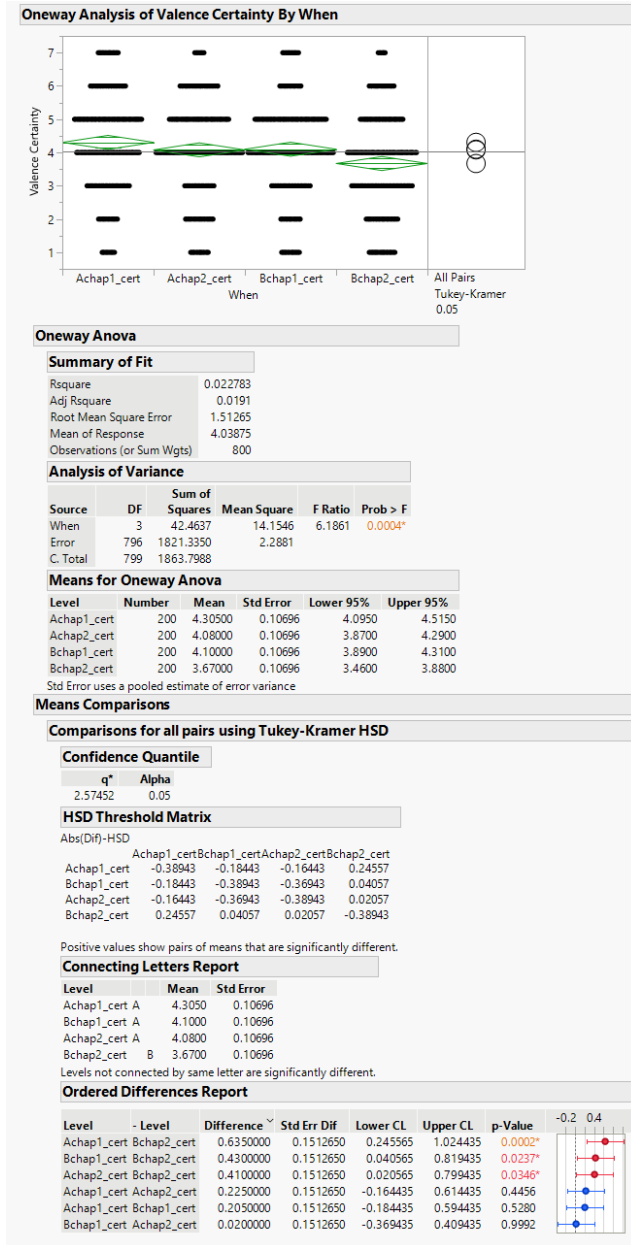


Figure A4

ANOVA reports for arousal certainty over chapter. The left report contains a means comparison over all chapters. The right report contains the mean comparison blocking for story. Both highlight a decline in user arousal certainty.

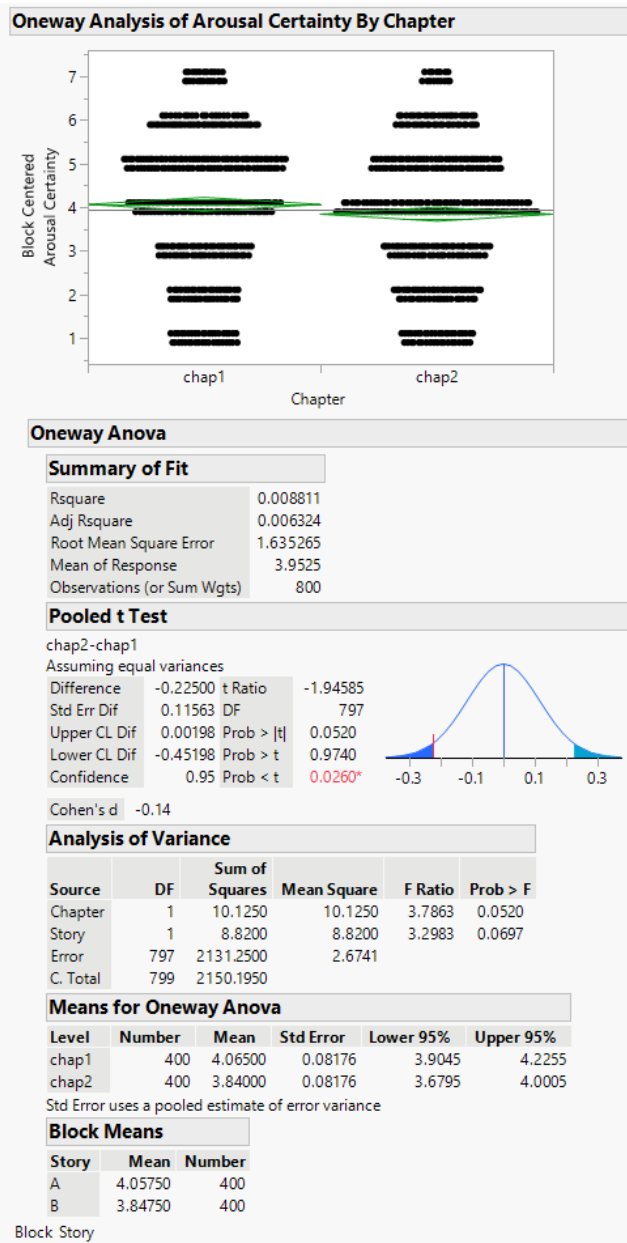
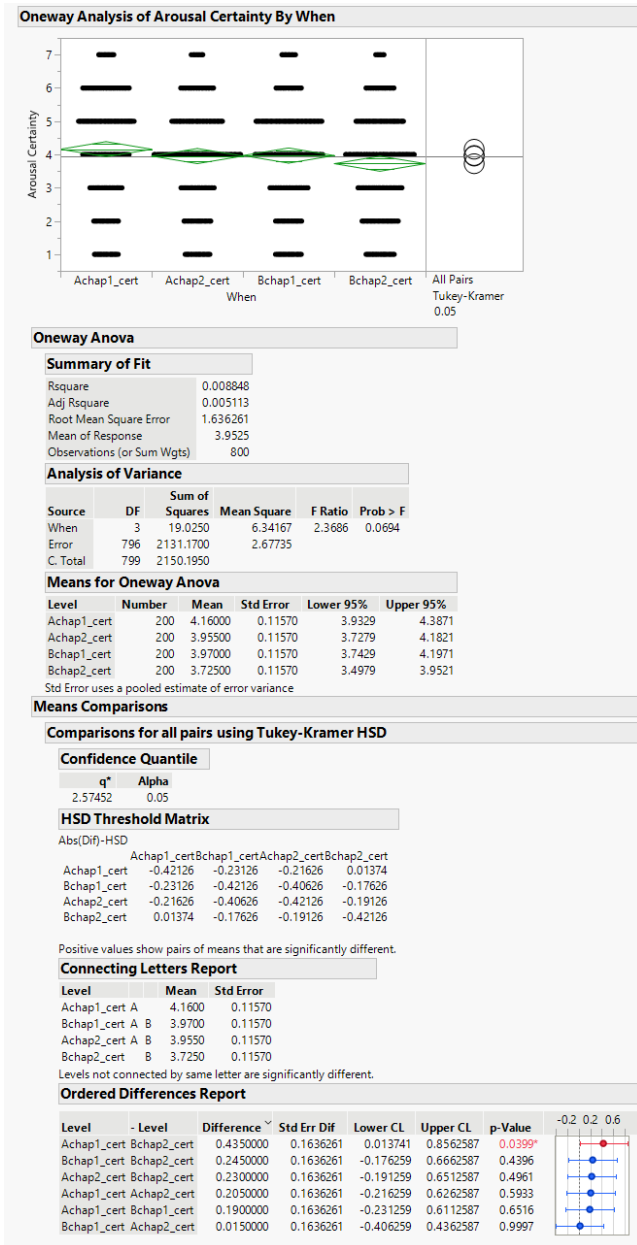
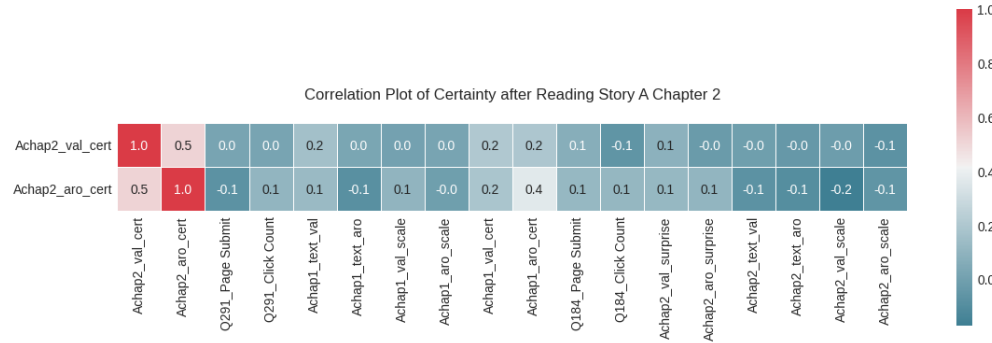
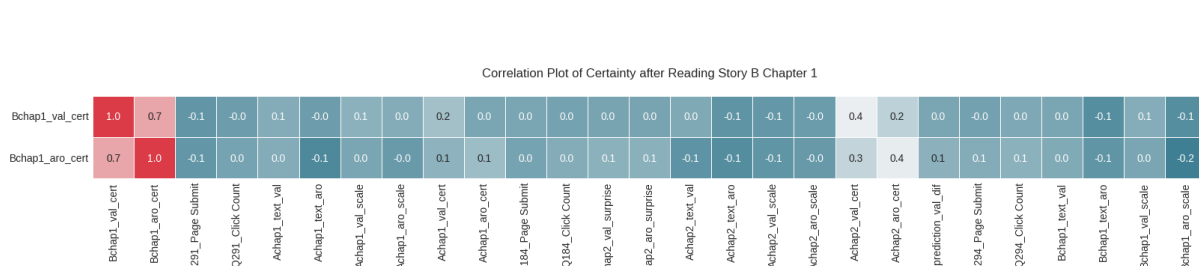


Figure A5*Correlation plots for valence and arousal certainty at each chapter throughout the survey.*

Correlation Plot of Certainty after Reading Story A Chapter 2



Correlation Plot of Certainty after Reading Story B Chapter 1



Correlation Plot of Certainty after Reading Story B Chapter 2

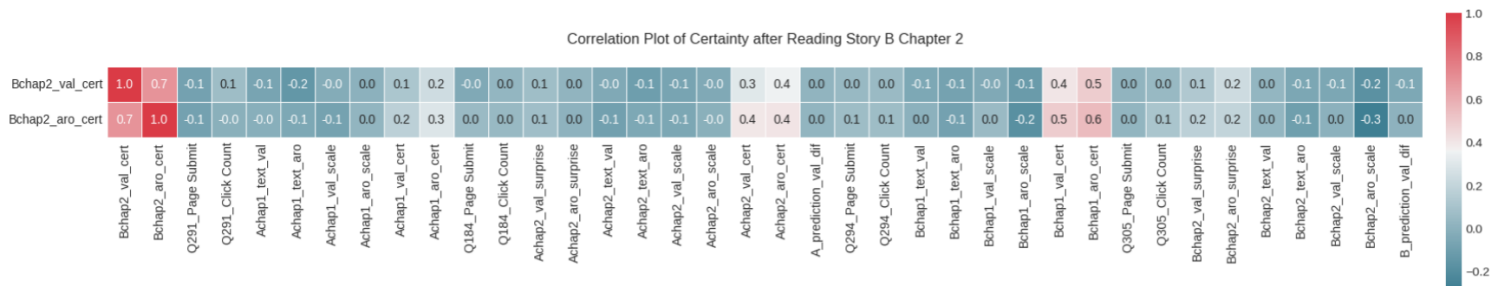


Figure A6

Stepwise regression report for valence certainty over chapter. Simple effect summary table provided.

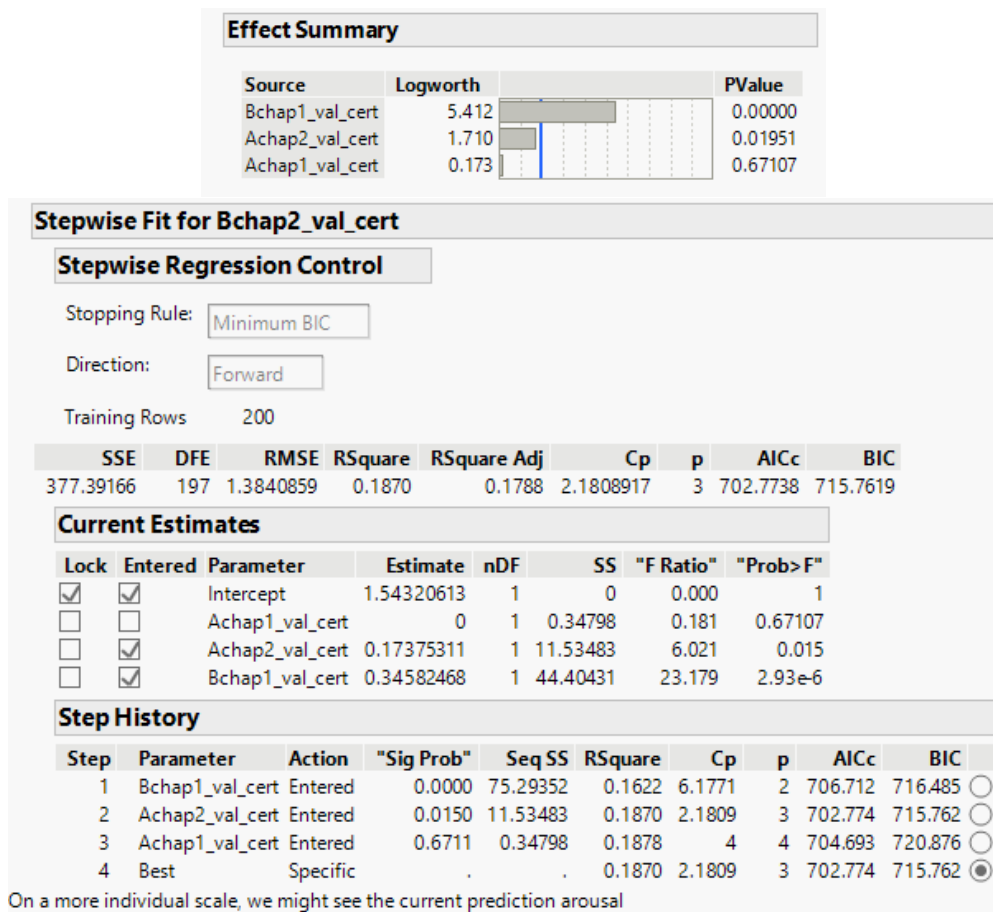


Figure A7

Stepwise regression report for arousal certainty over chapter. Simple effect summary table provided.

